

B.E. (COMPUTER SYSTEMS)
COURSES OF STUDIES (BATCH 2025 PROSPECTUS IN 2018 SIDE-BY-SIDE FORMAT)

FIRST YEAR

Fall Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
EF-101	IT Fundamentals and Applications	2	1	3
CS-115	Computer Programming	3	1	4
MT-116	Calculus & Analytical Geometry	3	0	3
EA-128	Functional English	3	0	3
ES-105 / ES-127	Pakistan Studies OR Pakistan Studies (For Foreigners)	2	0	2
PH-129	Applied Physics	3	1	4
CY-100	Essentials of Chemistry <i>(For Computer Science only)</i>	3	0	NC

Spring Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-117	Discrete Structures	3	0	3
CS-116	Object Oriented Programming	3	1	4
EE-125	Basic Electrical Engineering	3	1	4
ES-108	Ideology and Constitution of Pakistan	2	0	2
ES-206 / ES-209	Islamic Studies OR Ethical Behaviour	2	0	2
MT-221	Linear Algebra & Ordinary Differential Equations	3	0	3

SECOND YEAR

Fall Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-220	Digital Logic Design	3	1	4
CS-218	Data Structures & Algorithms	3	1	4
EA-218	Business Communication	2	1	3
EL-106	Basic Electronics	3	1	4
EF-309	Occupational Health and Safety	2	0	2
CS-253	Sustainable Engineering Practices	2	0	2

Spring Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-221	Computer Organization & Design	3	1	4
CS-215	Signals and Systems	2	1	3
CS-222	Database Management Systems	3	1	4
CS-326	Software Engineering	3	1	4
CS-219	Computer Engineering Workshop	0	1	1
EF-201 / EF-200	Civics & Community Engagement OR Community Service	2/0	0/0	2/NC

THIRD YEAR

Fall Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-327	Computer Communication Networks	3	1	4
CS-328	Computer Architecture	3	0	3
CS-373	Artificial Intelligence & Machine Learning	3	1	4
MT-335	Probability & Statistics	3	0	3
CS-419	Digital Signal Processing	3	1	4

Spring Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-329	Operating Systems	3	1	4
CS-301	Microprocessor Based System Design	3	1	4
CS-34X	Elective-I (AI Domain)	3	1	4
MT-442	Numerical Methods	3	0	3
EF-305	Engineering Economics & Management	3	0	3
EA/ES-###	Foreign Language I	2	0	NC

FINAL YEAR

Fall Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-406	Computer Engineering Design Project *	0	3	3
CS-431	Digital System Design	3	1	4
CS-36X	Elective-II (Network/Cyberspace Domain)	2	0	2
CS-471	Modeling and Simulation	3	0	3
CS-432	Distributed Computing	2	1	3
EA/ES-###	Foreign Language	2	0	2

Spring Semester				
Course Code	Course Title	Credit Hrs		
		Th	Pr	Total
CS-406	Computer Engineering Design Project *	0	3	3
CS-45X	Elective-III (System Design Domain)	2	1	3
CS-46X	Elective-IV (Software Engineering Domain)	2	1	3
CS-472	Entrepreneurship for Computer Engineers	2	0	2
CS-371	Project Management	2	0	2

* Duration is one academic year. Includes literature survey and initial work in the Fall Semester, with final examination and evaluation evaluated at the end of the Spring Semester.